

CHAMPS' Contribution to the Eighth Drag Prediction Workshop Buffet Working Group

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Introduction

Works within the DPW8/AePW4 Buffet Working Group

Results for test case 1:

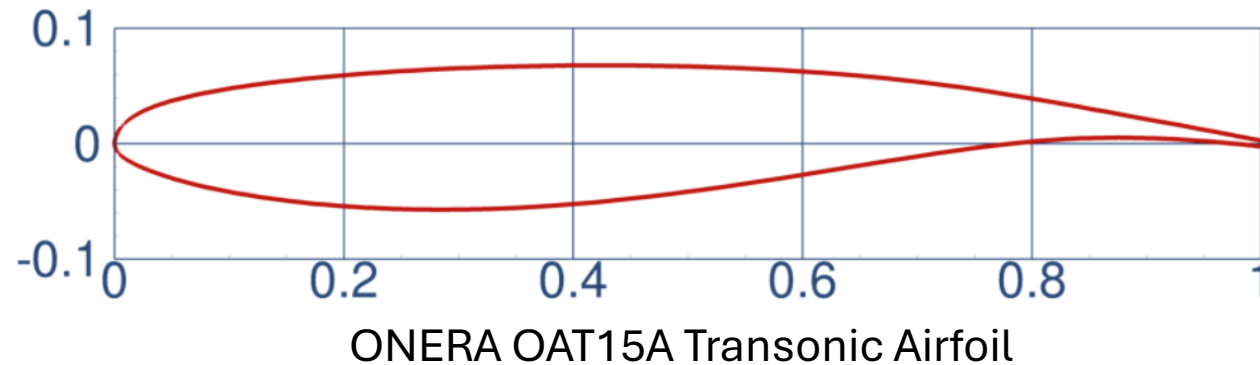
$Ma=0.73$, $Re=3M$, $T=271K$

Ranges of angles of attack : $1.36^\circ - 3.90^\circ$

Test case 1.a : RANS solutions

Test case 1.b : unsteady RANS solutions

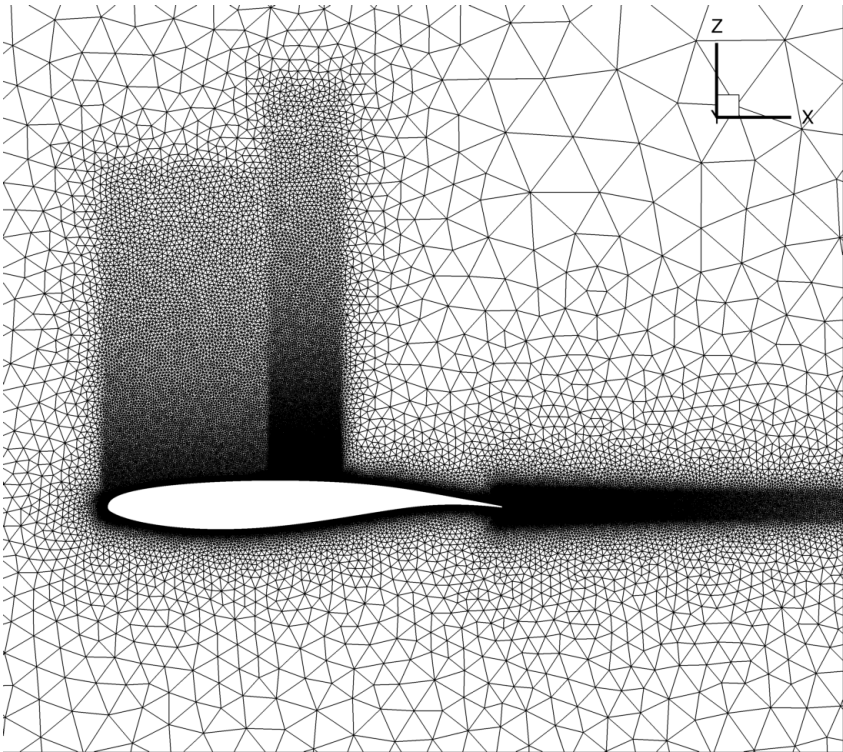
Jaquin, et al. "Experimental Study of Shock Oscillation over a Transonic Supercritical Profiles." AIAA Journal, Vol. 47, No. 9, 2009. Pages 1985-1994.



Grid

Committee-supplied Cadence Rev01 unstructured grids

Grid Level	Cell count	Target y+
L1	47,187	1.000
L2	89,616	0.670
L3	150,333	0.500
L4	235,491	0.400
L5	353,725	0.330
L6	517,448	0.290



ONERA OATA15A airfoil. Mesh Level 4.

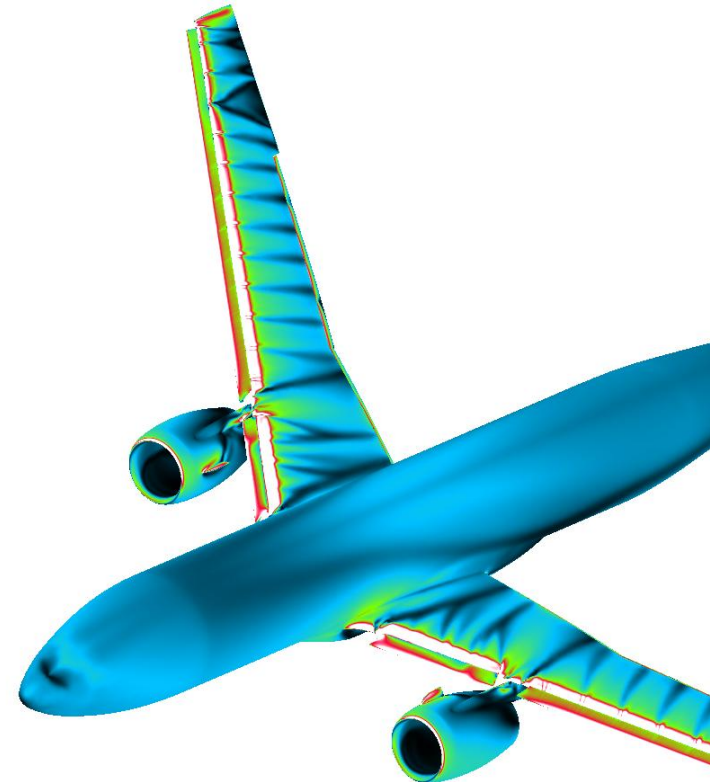


TC1.a Methodology

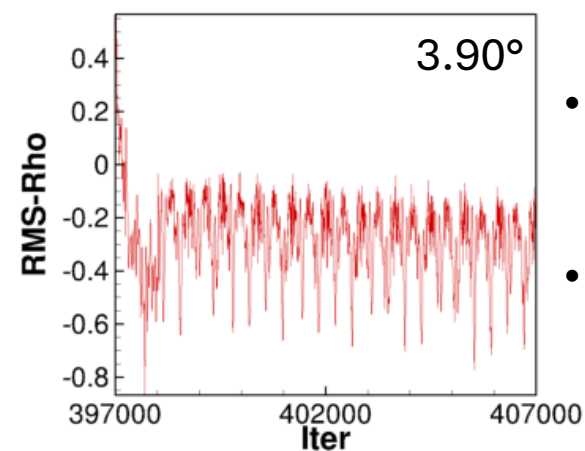
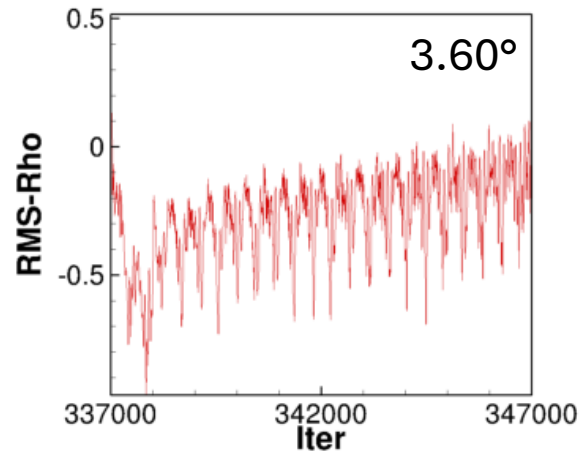
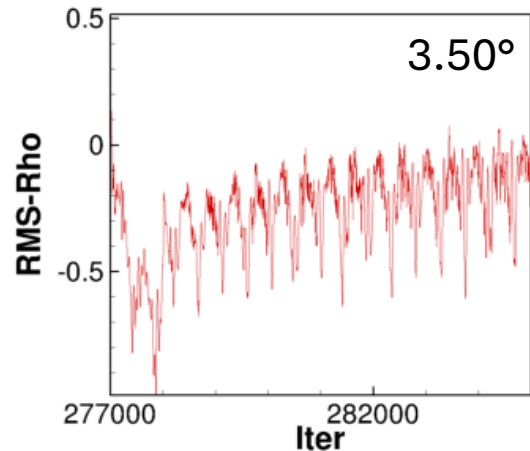
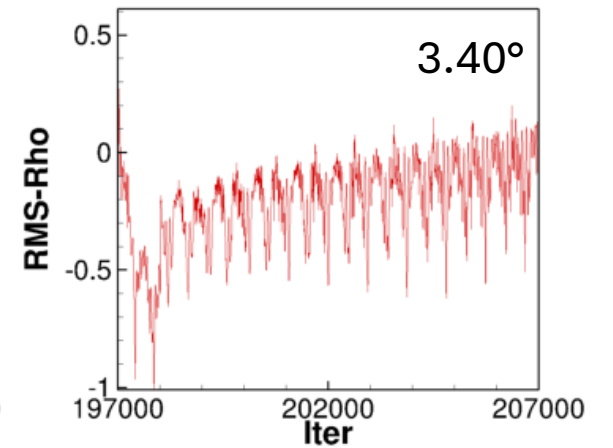
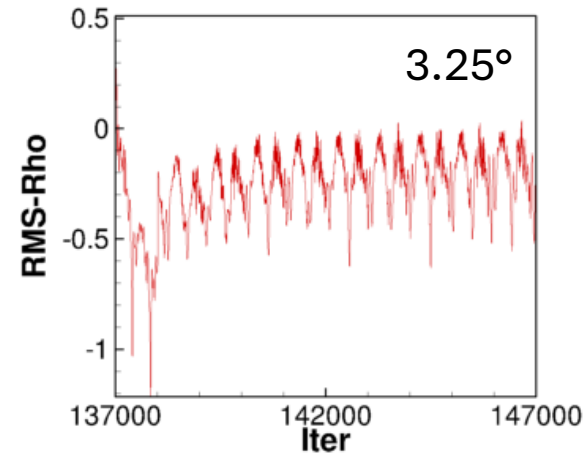
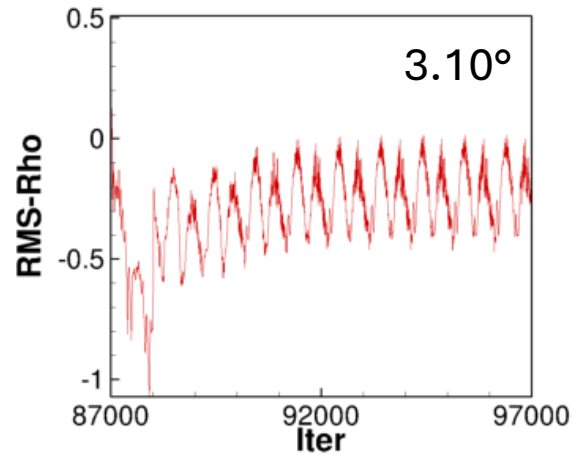
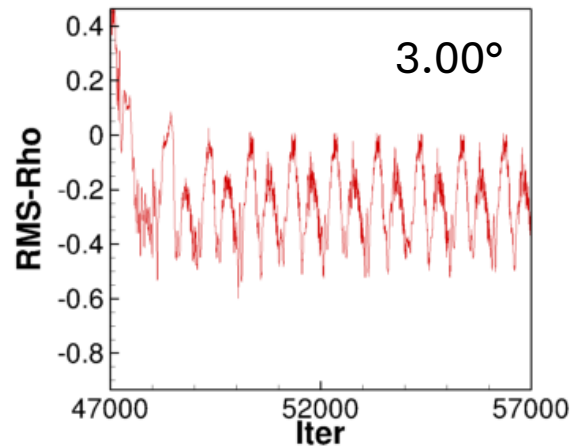
Flow solver: CHAMPS (CHApel Multi-Physics Simulation) first CFD solver developed using the Chapel programming language (Cray, HPE):

- FV - cell centered
- second-order Roe convective scheme
- Weighted-Least-Square gradients
- Nishikawa R5 gradient limiter [1] ramped down and finally disabled
- SA turbulence model
- Block-SGS implicit scheme
- Usage of Selective Frequency Damping (SFD) to enable deep convergence of residuals

Warm starts from previous angle of attack.



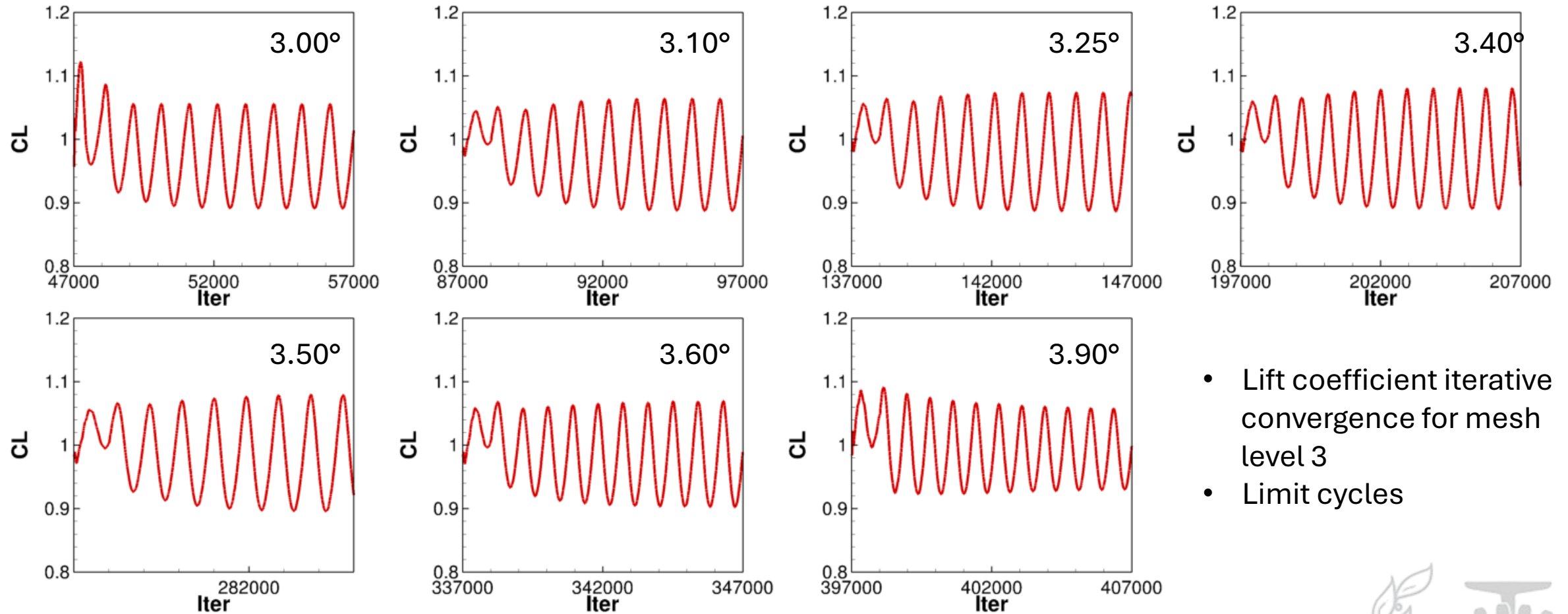
TC1.a Results



- Mass-conservation residuals iterative convergence for mesh level 3
- Limit cycles



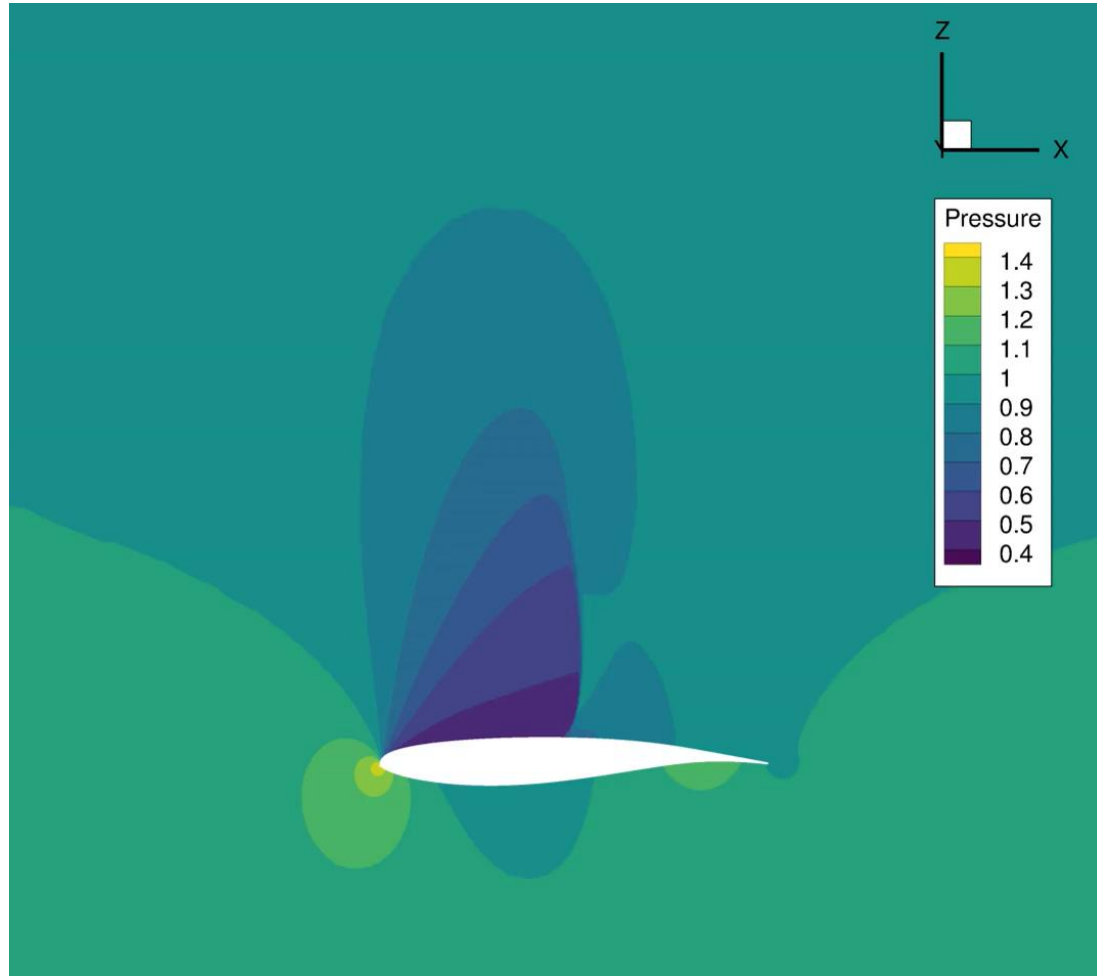
TC1.a Results



- Lift coefficient iterative convergence for mesh level 3
- Limit cycles



TC1.a Results



Extract the flow field over 3 pseudo periods at 3.90° (mesh L3):

- Pseudo unsteadiness with local time stepping
- Steady state is not reached as the solver faces unsteady mechanism (expected as it is a buffet configuration)



Selective Frequency Damping

Steady-state RANS flow solvers encounter convergence problems when trying to solve flow conditions with unsteady phenomena (buffet or vortex-shedding). However, a fixed point steady-state solution can exist in these conditions, which is useful in many engineering applications such as design or stability analysis.

The Selective Frequency Damping (SFD) developed by Åkervik [2] and Jordi [3] (Implemented in CHAMPS by Liguori et al.[4]) method aims to stabilize such unstable flows. Original solver equation:

$$\dot{q} = f(q)$$

Under the assumption that a steady state q_s exists:

$$\dot{q}_s = f(q_s) = 0$$



Selective Frequency Damping

Under the assumption that a steady state exists: q_s

$$\dot{q}_s = f(q_s) = 0$$

Augmented SFD equation: Add source term which is equivalent to a proportional controller (eq. 3) where χ is the control coefficient

$$\dot{q} = f(q) - \chi(q - q_s) \quad (2)$$

However, q_s is not known a priori and is then replaced by $\bar{q} = T * q$ a low-pass filtered version of the flow (convolution of q with a temporal filter kernel T).

$$\dot{q} = f(q) - \chi(q - \bar{q}) \quad (3)$$

As q reaches q_s , $\bar{q}$ also reaches q_s . For a steady solution, the source term will tend toward zero: **no modification of the solution:**

$$\dot{\bar{q}} = \frac{q - \bar{q}}{\Delta} \quad (3)$$

With $\bar{q}$ the low-pass filtered version of the flow.

χ equivalent to a proportional gain and Δ the cutoff wavelength of the filter



Selective Frequency Damping

The a low-pass filtered version of q is written as

$$\bar{q}(t) = \int_{-\infty}^t \frac{1}{\Delta} \exp(\tau - t) q(\tau) d\tau$$

With Δ the filter width. This formulation is impractical as it would require to save the entire history of q . Therefore, the equivalent formulations is used:

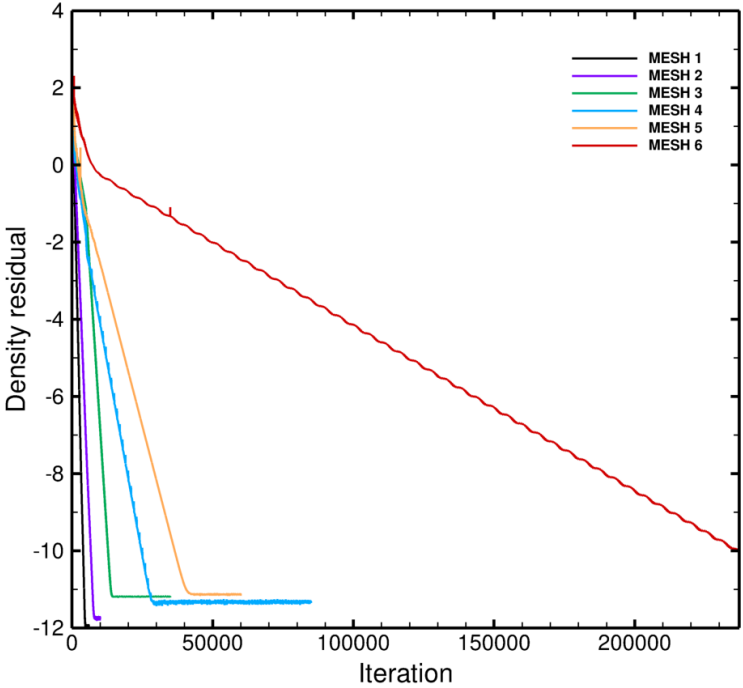
$$\dot{\bar{q}} = \frac{q - \bar{q}}{\Delta} \quad (3)$$

The final version of the system is:

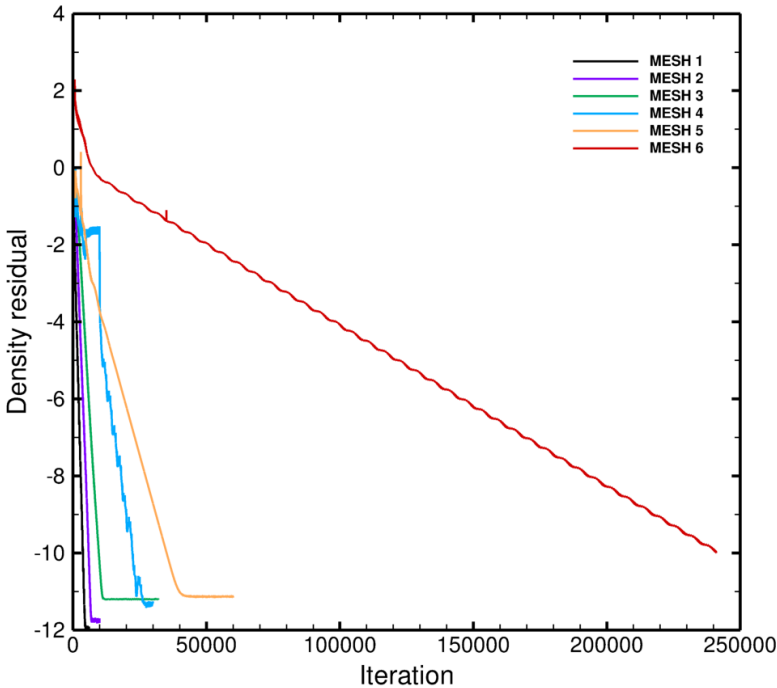
$$\begin{cases} \dot{q} = f(q) - \chi(q - \bar{q}) \\ \dot{\bar{q}} = \frac{q - \bar{q}}{\Delta} \end{cases}$$



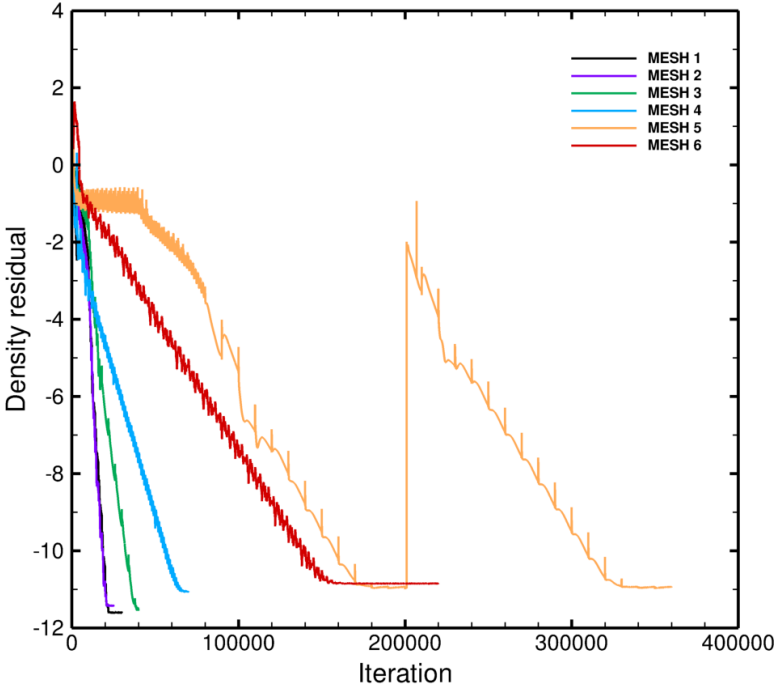
TC1.a Results



1.36°



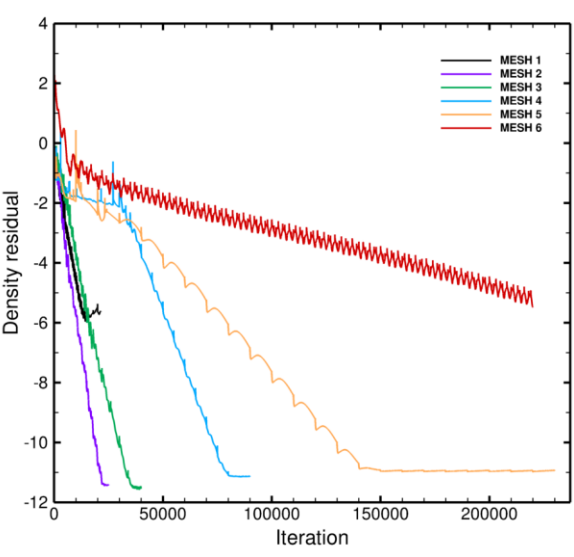
1.50°



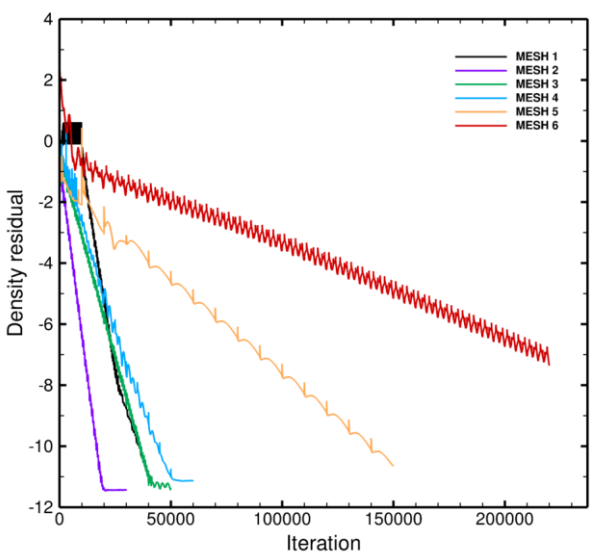
2.50°



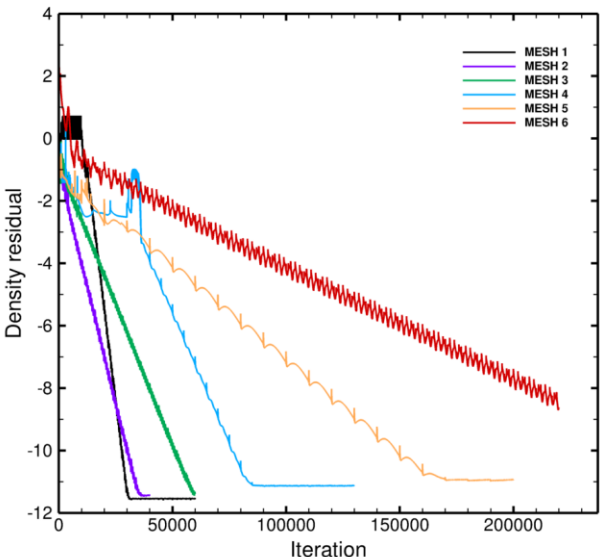
TC1.a Results



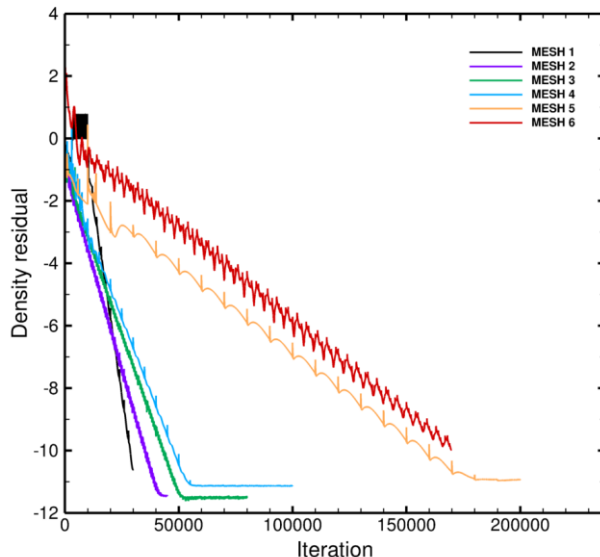
3.00°



3.10°



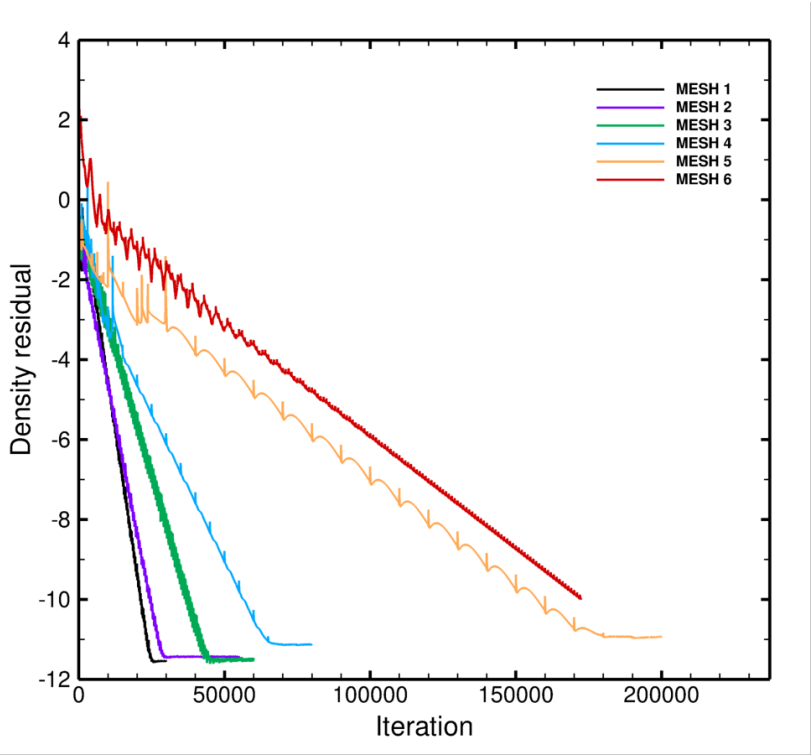
3.25°



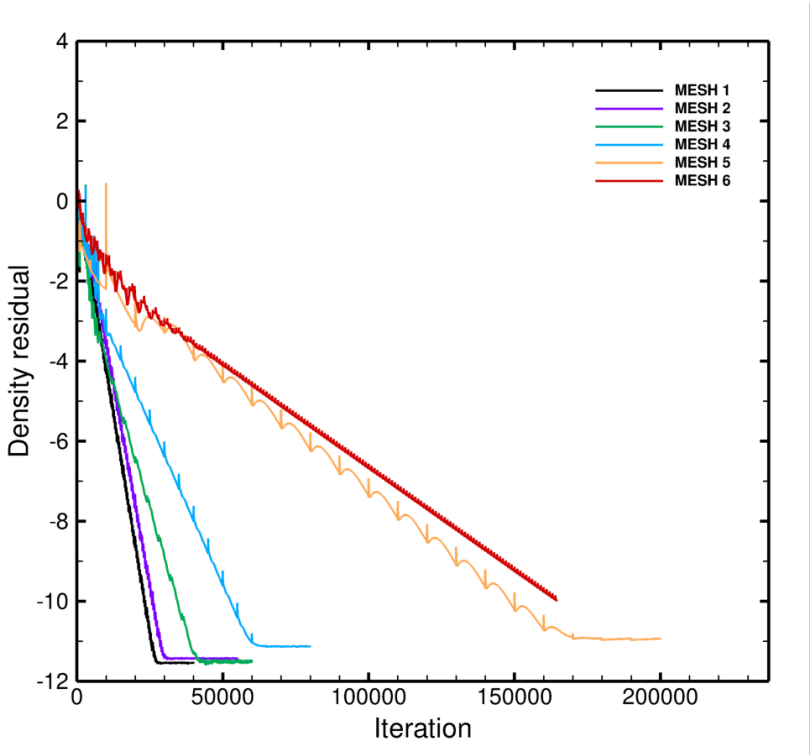
3.40°



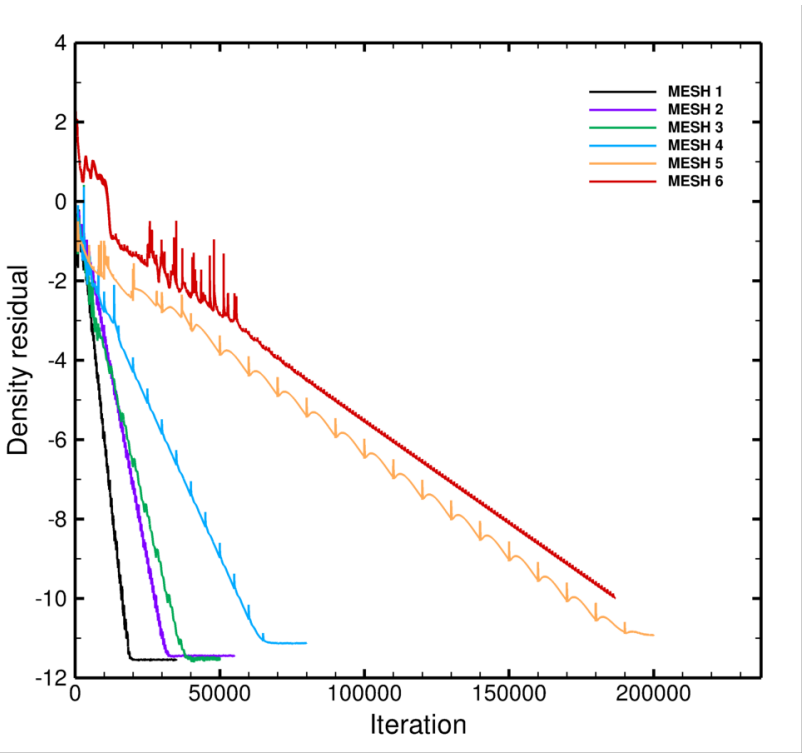
TC1.a Results



3.50°



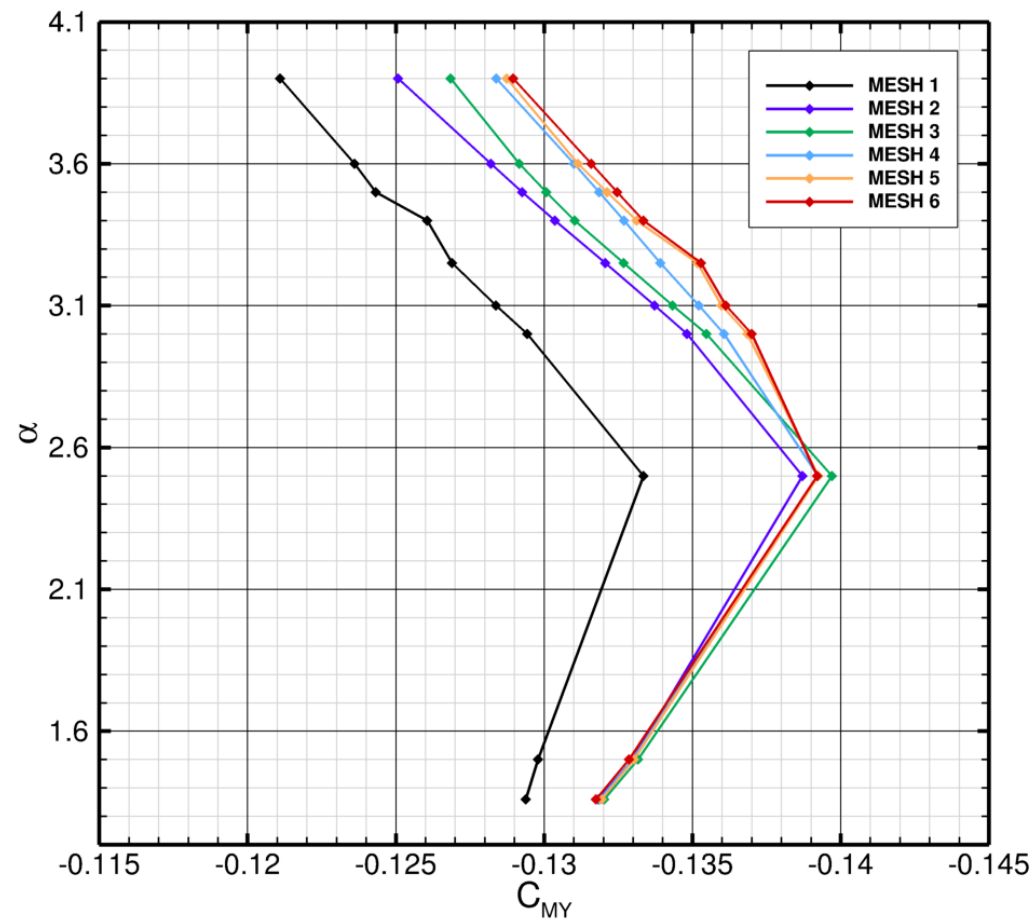
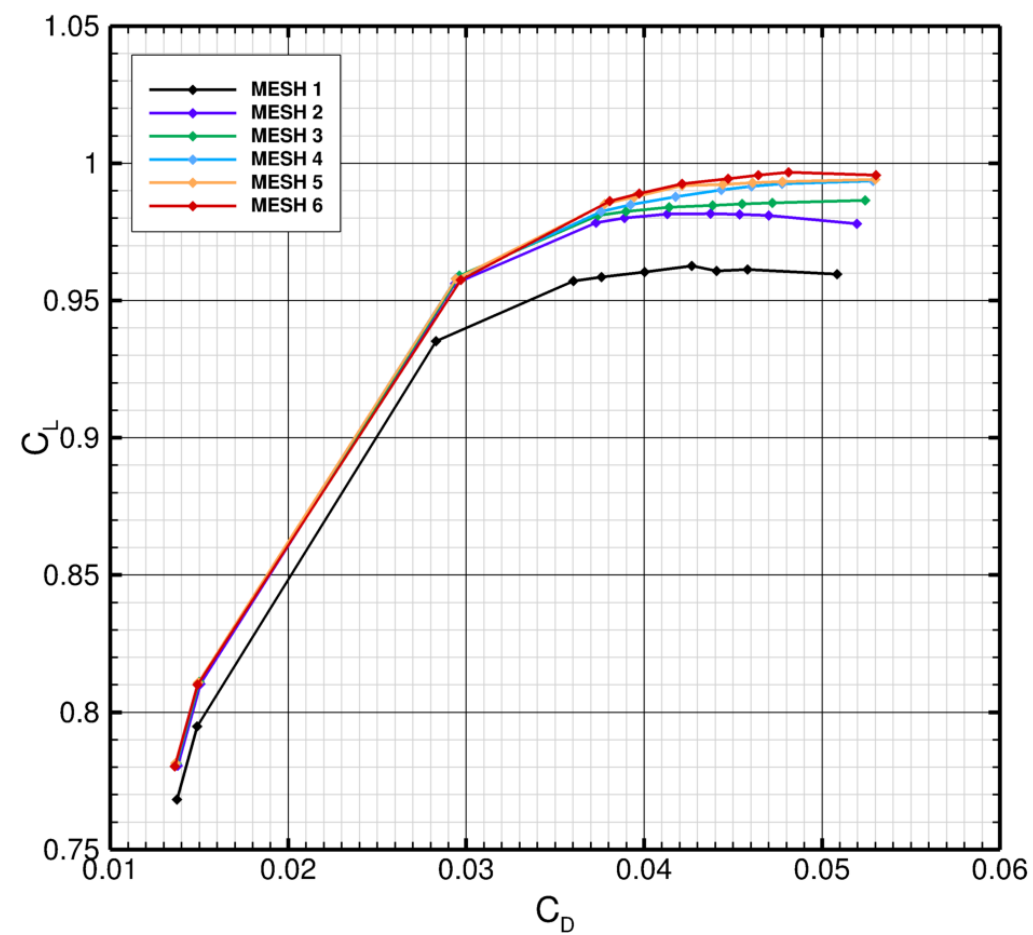
3.60°



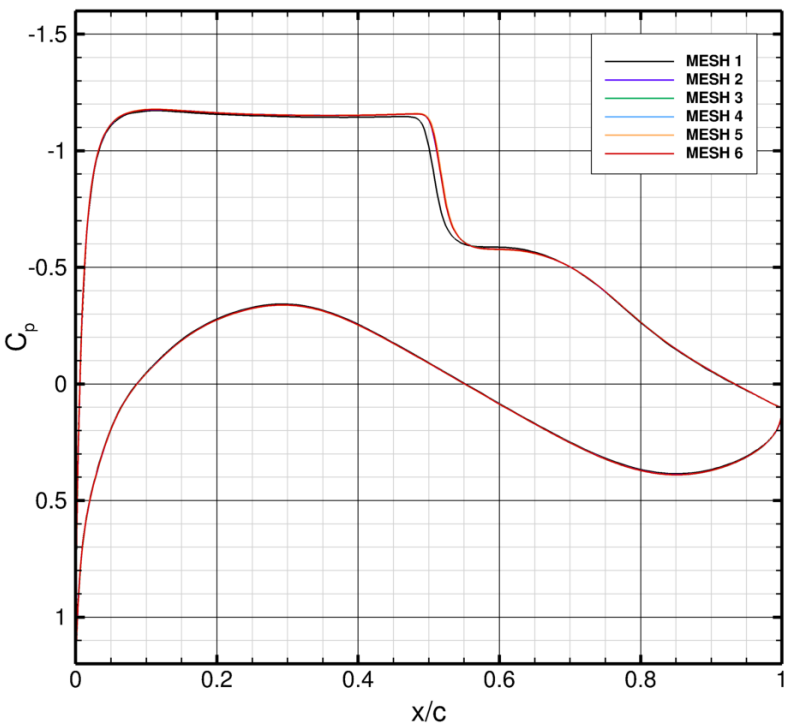
3.90°



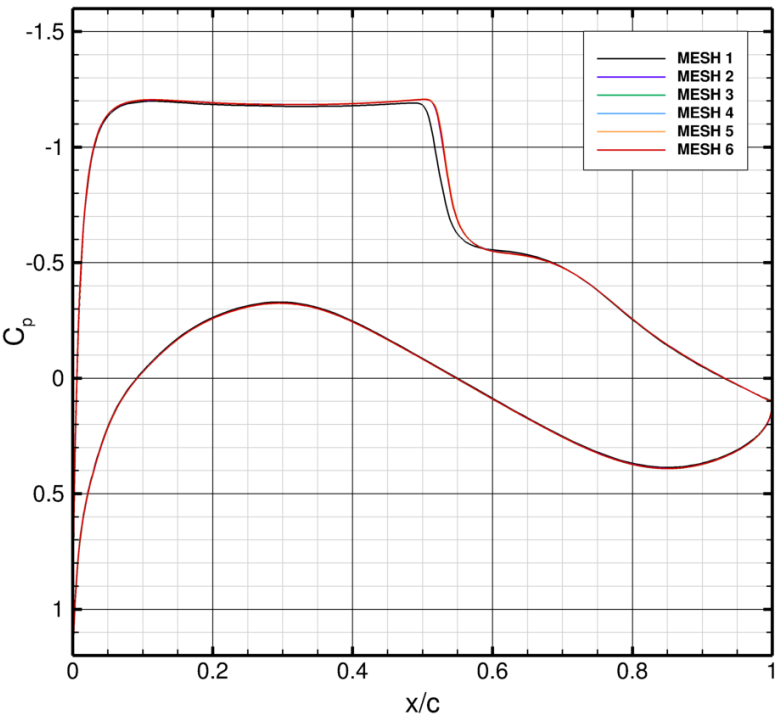
TC1.a Results



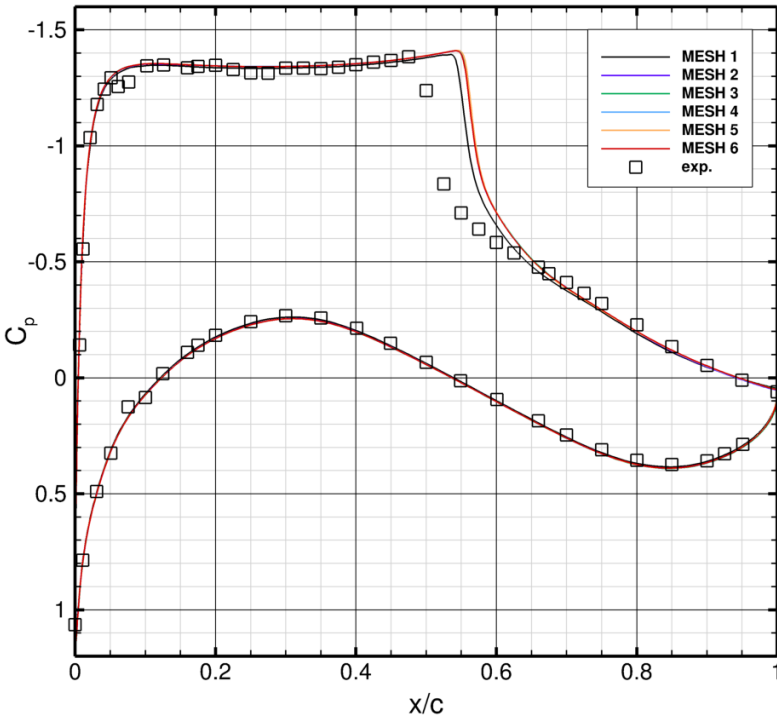
TC1.a Results



1.36°



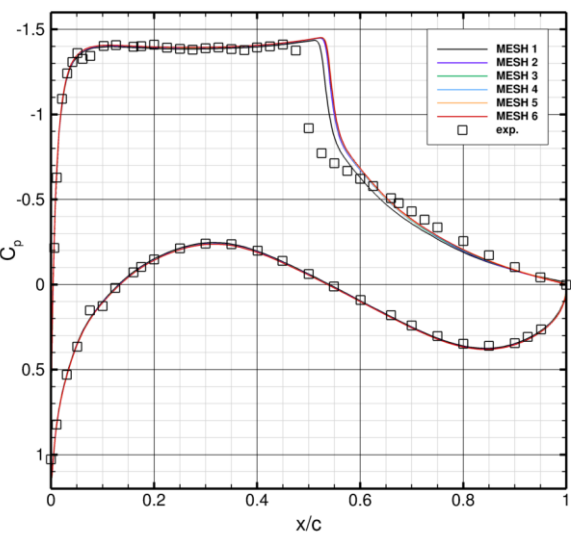
1.50°



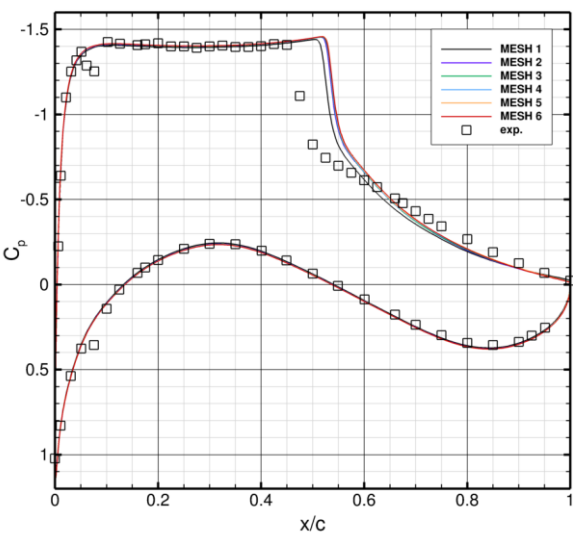
2.50°



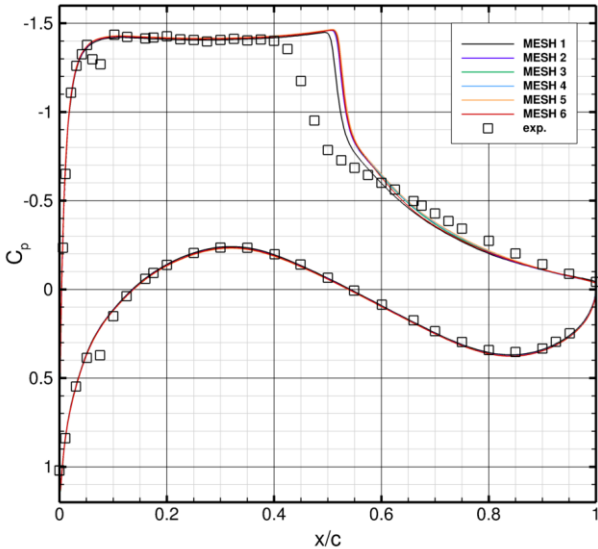
TC1.a Results



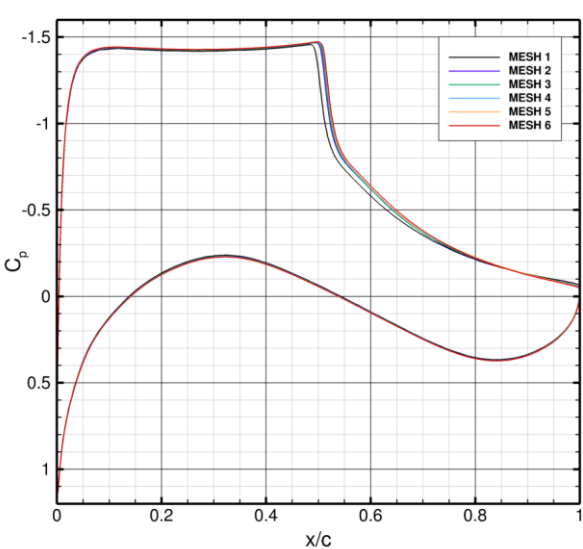
3.00°



3.10°



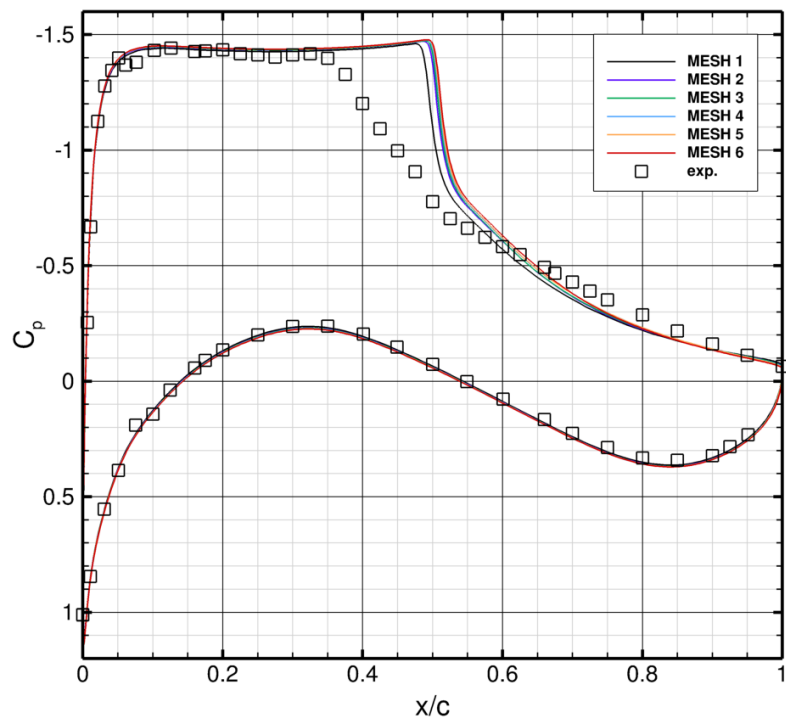
3.25°



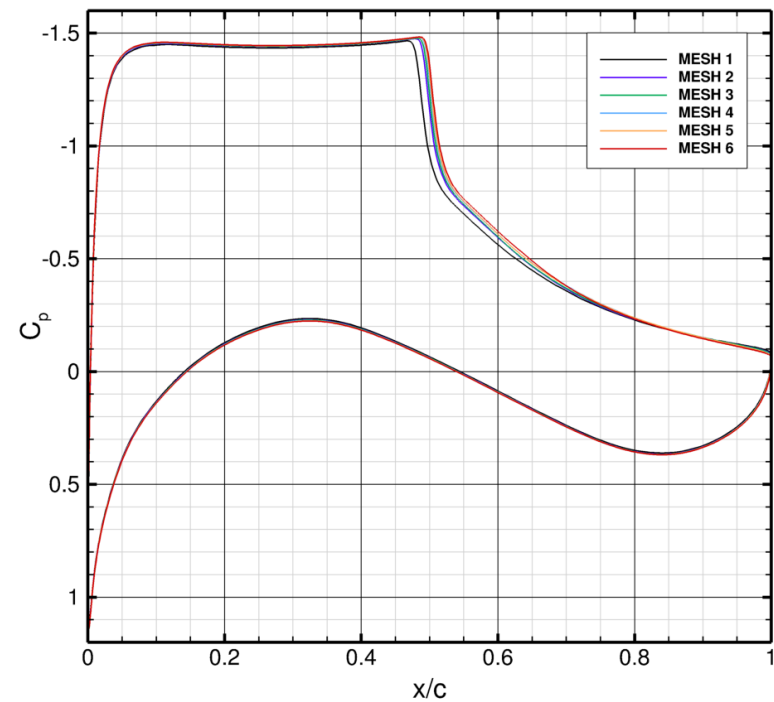
3.40°



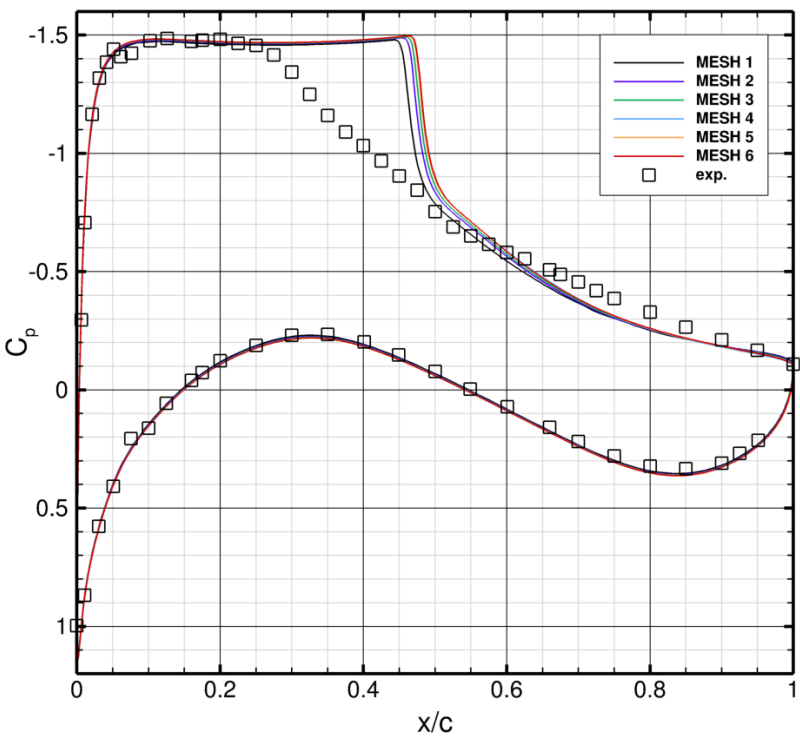
TC1.a Results



3.50°



3.60°



3.90°



TC1.b Methodology

Warm starts from RANS solutions

SA-comp model was used

Physical time step of 0.086 CTU

Simulations run through the transient phase until buffet cycles fully develop (~150CTU).

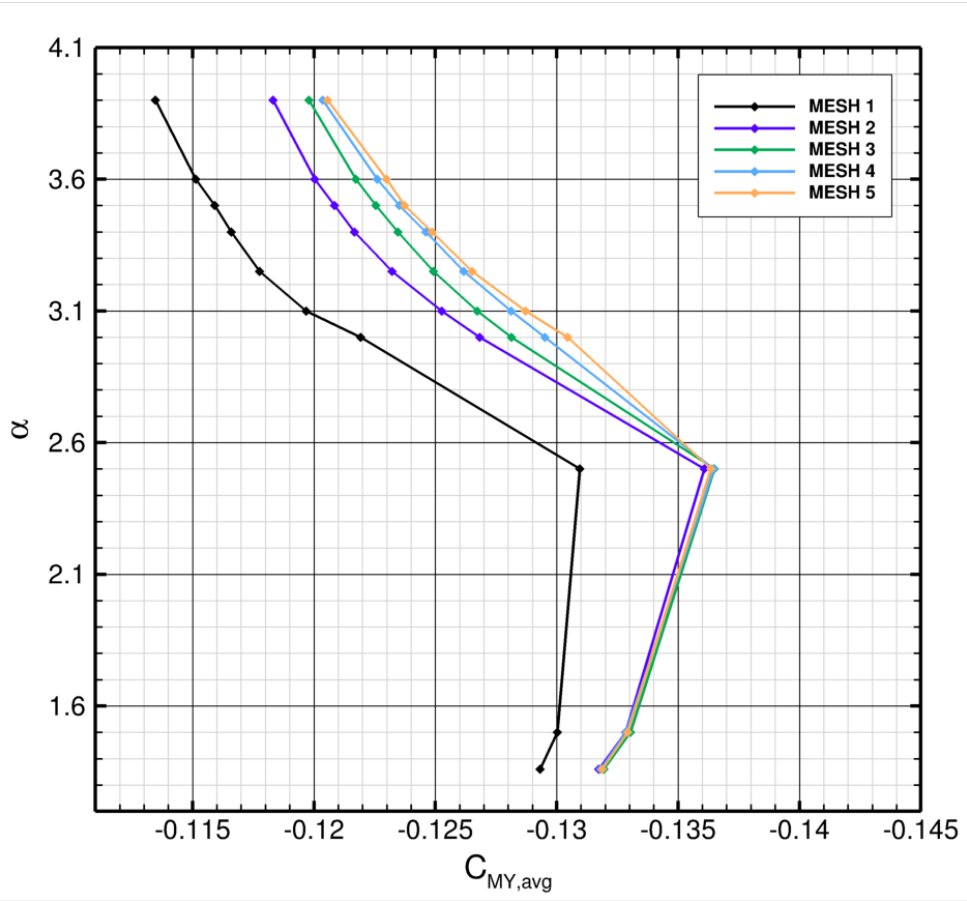
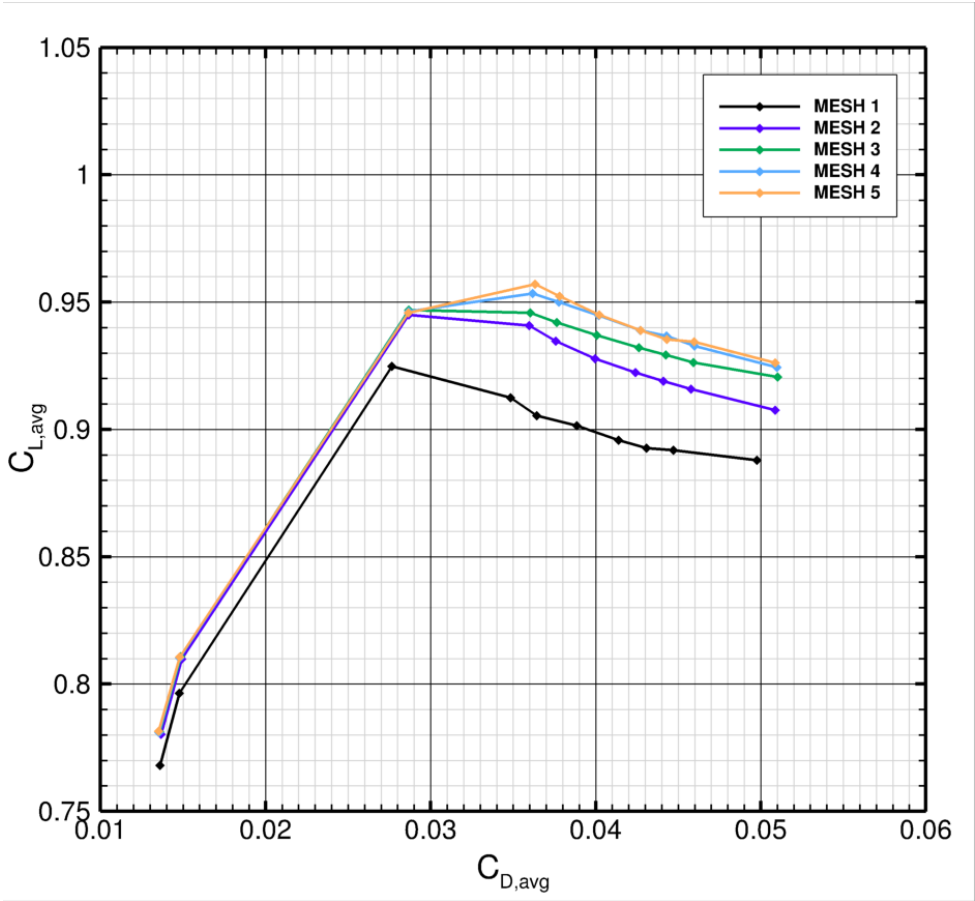
Time averaging over 10 buffet cycles (~150CTU)

Nishikawa Limiter [1] was used with the minimum possible amount of dissipation to ensure stability.

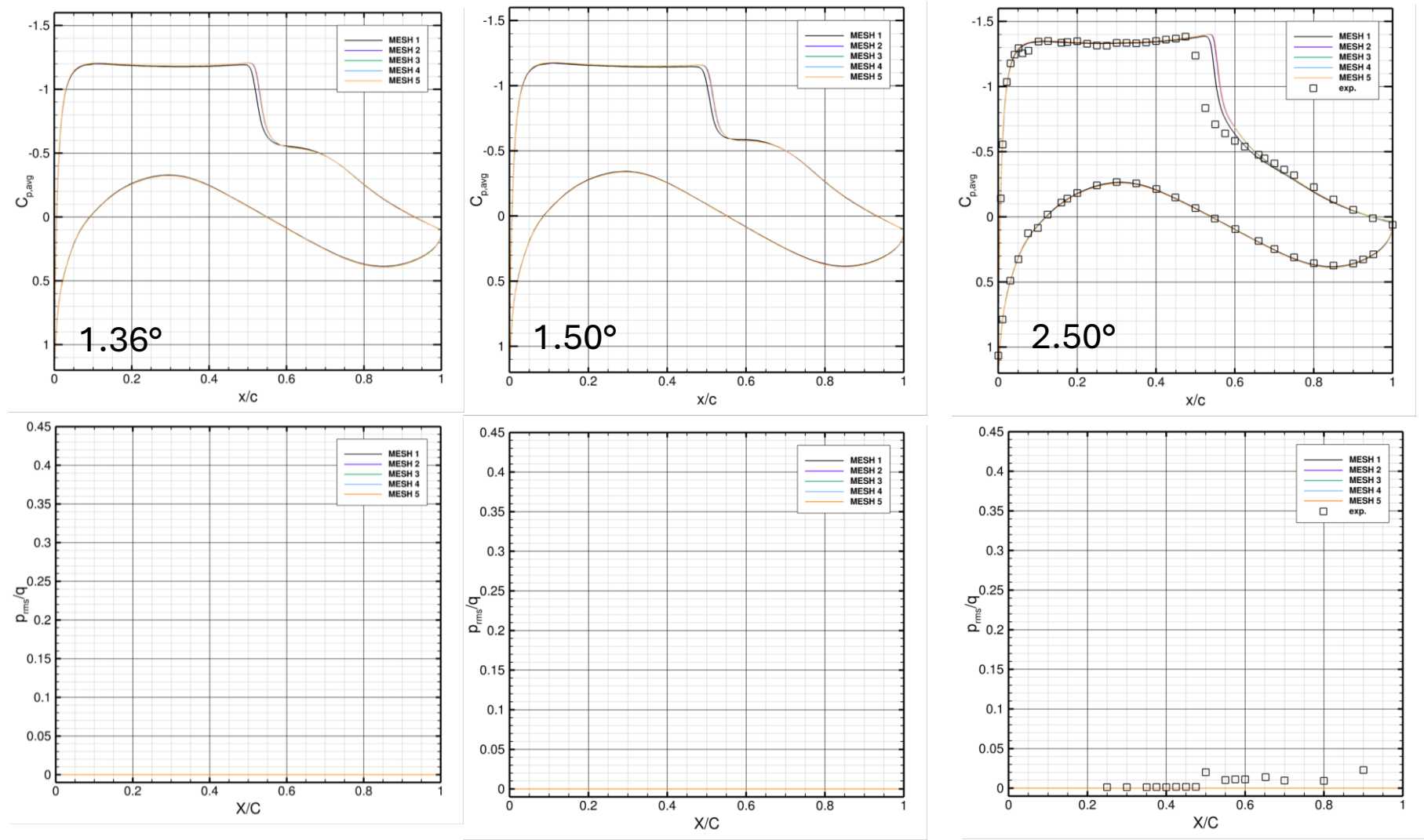
Meshes L1, L2, L3, L4, L5 only were used for simulations.



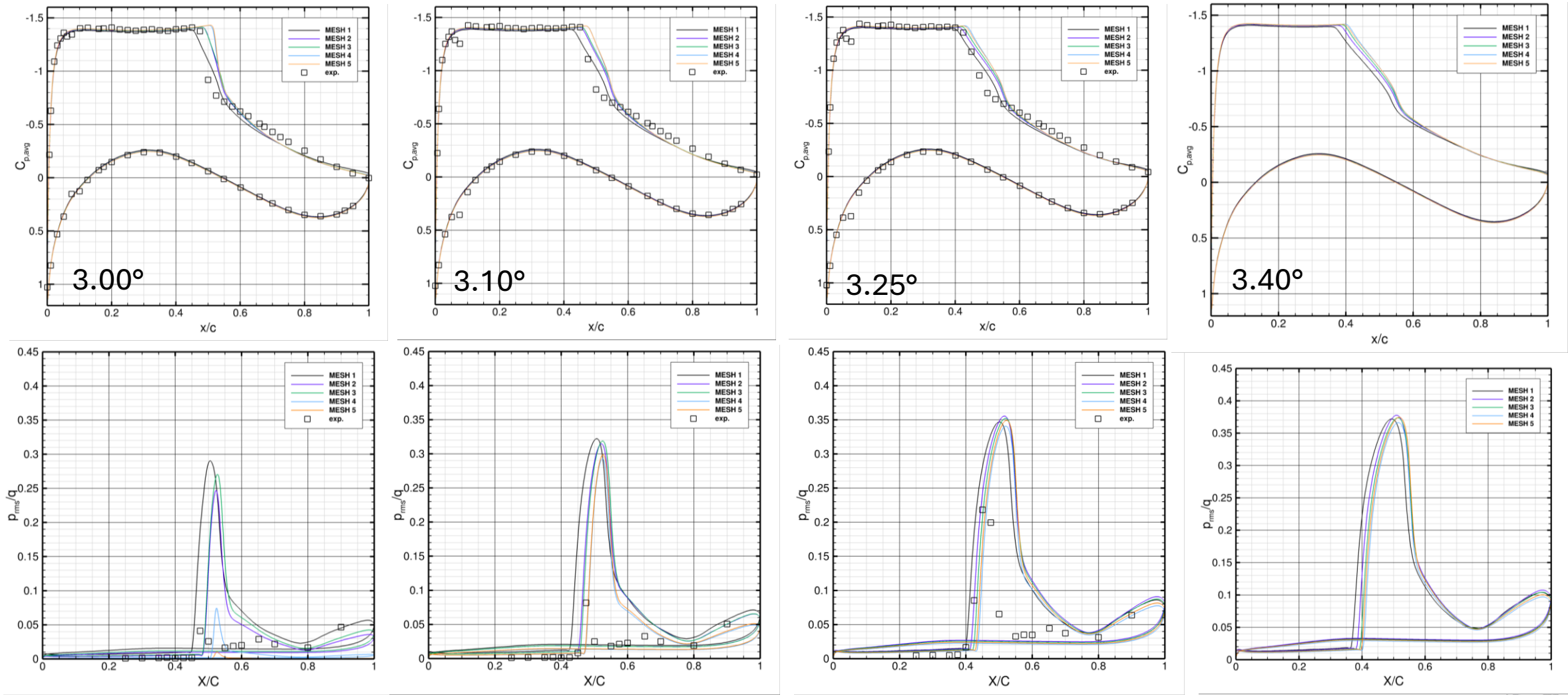
TC1.b Results



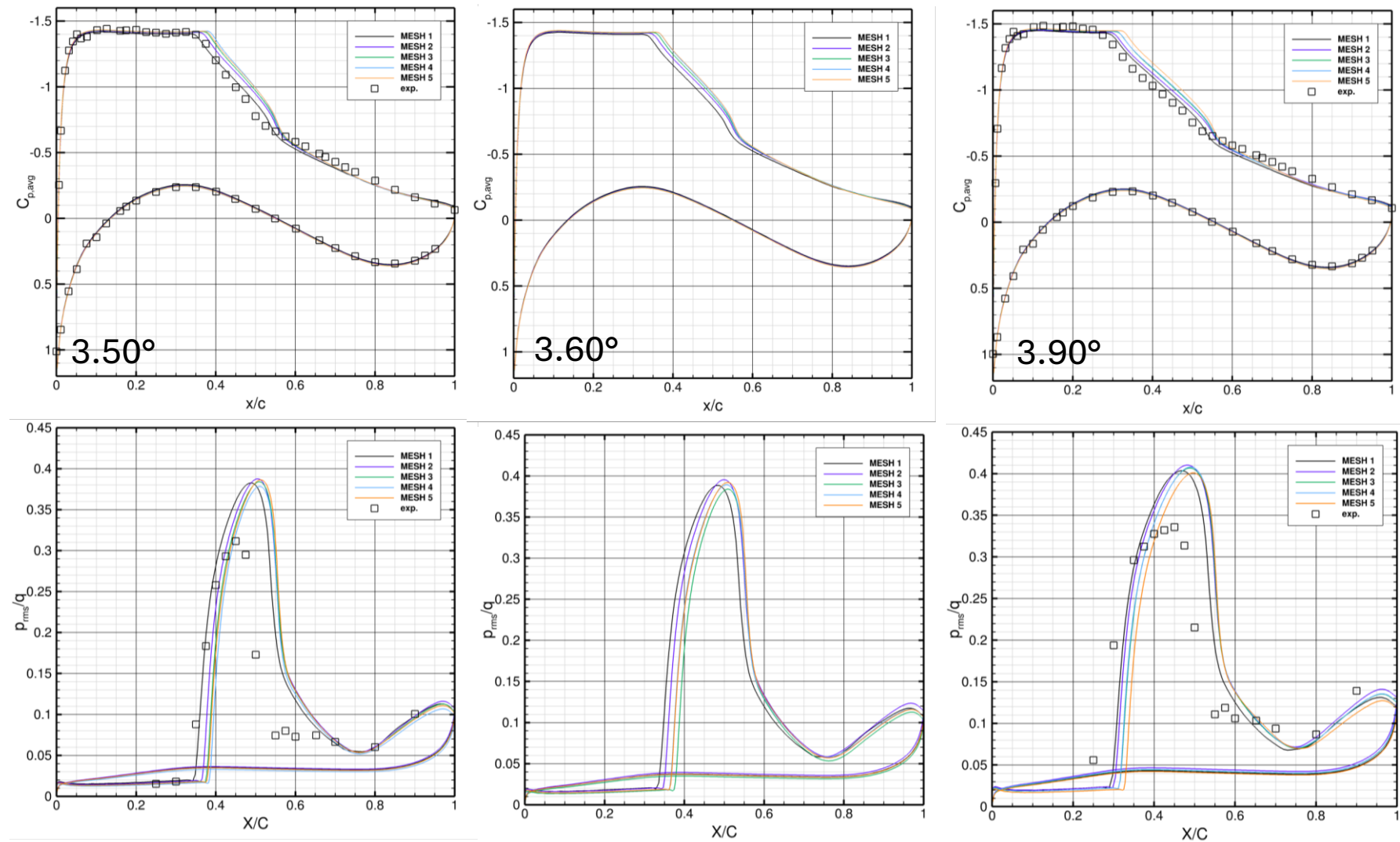
TC1.b Results



TC1.b Results



TC1.b Results



For all simulations, a Strouhal Number of 0.069 was obtained Consistent with work from Plant, F. [5]



Conclusion

Deep iterative convergence of residuals was achieved using Selective Frequency Damping (SFD), isolating remaining discrepancies to modeling or grid-related errors.

Unsteady solutions with URANS show significantly improved agreement with experimental data. However, differences persist, notably in the pressure coefficient downstream of the shock.

The RMS pressure peak is not accurately captured, despite a good match in its location.

Further improvements will require scale-resolving simulations (e.g., DES) to better capture the unsteady flow features.



References

- [1] H. Nishikawa. New unstructured-grid limiter functions. In SciTech 2022 Forum, AIAA Paper 2022-1374, San Diego, CA
- [2] E. Åkervik, L. Brandt, D. S. Henningson, J. Hoepffner, O. Marxen, and P. Schlatter, “Steady solutions of the navier-stokes equations by selective frequency damping,” *Physics of fluids*, vol. 18, no. 6, 2006.
- [3] B. E. Jordi, C. J. Cotter, and S. J. Sherwin, “Encapsulated formulation of the selective frequency damping method,” *Physics of Fluids*, vol. 26, no. 3, 2014.
- [4] V. Liguori, F. Plante, and E. Laurendeau, “Implementation of an efficient selective frequency damping method in a rans solver,” in *AIAA Scitech 2021 Forum*, 2021, p.0359.
- [5] Plante, F. (2020). *Towards understanding stall cells and transonic buffet cells*. Ecole Polytechnique, Montreal (Canada).



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